

**AMENDMENT NO. 1 JANUARY 2009
TO
IS 8188 : 1999 TREATMENT OF WATER FOR COOLING
TOWERS — CODE OF PRACTICE**

(First Revision)

[Page 3, clause 6.3.2.5(a)] — Substitute the following for the existing text

'Dispersants — Suitable chemicals in specified amounts are added as sludge fluidizers, surfactants and wetting agents. These keep particles suspended in the water and prevent deposit formation. Natural dispersants such as lignins and tannins provide good results but these must be used continuously and relatively at high dosages (50 to 200 ppm). However, these provide excellent food for biological organisms which cultivate and create another problem in the system. They also react with chlorine or iron salts. Hence, synthetic polymers most commonly polyacrylates/methacrylates of different molecular weights are used at concentration of 4 to 5 ppm. Dosage level is determined by considering the water characteristics, cycle of concentration, temperature profile on the process side, etc. Lower levels are used in once through cooling systems. Now-a-days, different types of tagged polymers are available which can be monitored on line to determine the optimum dosage level.'

[Page 4, clause 6.3.2.5(d)(v)] — Add the following after (v)'

- 'vi) *Dibromonirilapropionamide (DBNPA)* — This is used in large recirculating and once through cooling tower. DBNPA is the fastest active non-oxidizing biocide and is not persistent. This hydrolyses quickly under both acidic and alkaline conditions and hence increased dosages must be used. It is preferred for its instability in water as it quickly kills bacteria and then degrades to ammonia and bromide ions.
- vii) *Glutaraldehyde* — This versatile molecule is used to control the growth of bacteria, fungi and algae, including sulphate reducing bacteria. Glutaraldehyde is one of the few biocides that contains only carbon, hydrogen and oxygen, thus making it both halogen-free (no AOX issues) and readily biodegradable in the environment. The carbonyl groups of glutaraldehyde cross-link with the amino groups (NH₂) contained within the cell wall of microorganisms, and in effect, prevents the microbe from receiving nutrients or discarding waste products through its cell. Glutaraldehyde is compatible with hydrogen sulfide, and is very effective against anaerobic bacteria. However, glutaraldehyde reacts with ammonia and hence, it is best to avoid using the glutaraldehyde in ammonia leaks situations or in treatment of ammonia chillers'

(CHD 13)